

Convexity-Based Approximation and Weighted Two-Class Reduced-Form Model

1 Core Idea: Convexity Approximation in the Non-Stochastic Model

In the original N-worker model, the lottery delivers a stochastic prize drawn from a Pareto distribution with heavy right tail. Low-position workers favor it because the utility function is convex below the reference point r ($\gamma_{\text{low}} > 1$), implying

$$\mathbb{E}[u(y)] > u(\mathbb{E}[y])$$

for random y with positive variance. Convexity amplifies rare large upside outcomes, creating effective risk-loving behavior despite a roughly mean-preserving lottery.

In the simplified non-stochastic two-class model, we avoid repeated random draws. Instead, we replace stochastic lottery prizes with a deterministic equivalent that includes a linearized risk premium capturing the second-order welfare effect of variance via utility curvature.

The effective lottery benefit for a representative agent in class w is

$$s_w \cdot \kappa \Pi + \rho_w,$$

where $s_w \cdot \kappa \Pi$ is the mean pool return share and

$$\rho_w = \frac{1}{2} u''(m_w) \cdot \hat{\sigma}_w^2$$

is the risk-adjustment term. A simple proxy for virtual variance is

$$\hat{\sigma}_w^2 = c \cdot ((1 - \lambda_w) m_w)^2 \quad (c > 0 \text{ calibration constant}).$$

With

$$u''(y) = \begin{cases} \gamma_{\text{low}}(\gamma_{\text{low}} - 1)y^{\gamma_{\text{low}}-2} & y \leq r \\ \gamma_{\text{high}}(\gamma_{\text{high}} - 1)y^{\gamma_{\text{high}}-2} & y > r \end{cases},$$

we obtain $u'' > 0$ below r ($\rho_w > 0$, bonus) and $u'' < 0$ above r ($\rho_w < 0$, penalty).

Advantages include deterministic dynamics, faster computation, and preservation of qualitative asymmetry. Limitations include loss of higher-order tail effects and dependence on calibration of c .

2 Weighted Two-Class Reduced-Form Model

We consider a reduced-form model with unequal population masses: large mass f_L of lower-ranked workers (typically $f_L \approx 0.8\text{--}0.95$) and small mass $f_H = 1 - f_L$ of higher-ranked workers. Aggregate

income shares remain $R\mu$ for the low class and $R(1 - \mu)$ for the high class ($\mu < 0.5$), so per-capita incomes are

$$m_L = \frac{R\mu}{f_L}, \quad m_H = \frac{R(1 - \mu)}{f_H}.$$

Each representative agent in class w chooses $\lambda_w \in [0, 1]$, the **fraction of per-capita income allocated to upskilling** (safe improvement channel). The fraction allocated to the shared lottery pool is therefore $1 - \lambda_w$.

Aggregate lottery pool:

$$\Pi = f_L(1 - \lambda_L)m_L + f_H(1 - \lambda_H)m_H.$$

Lottery return to a representative agent in class w :

$$\kappa f_w(1 - \lambda_w)m_w.$$

Net income per representative lower-ranked worker:

$$y_L = m_L + \eta\lambda_L m_L - C_L(\lambda_L) - (1 - \lambda_L)m_L \\ + \kappa f_L(1 - \lambda_L)m_L + \rho_L(1 - \lambda_L),$$

with $C_L(\lambda_L) \approx b\eta\lambda_L m_L$ and $\rho_L(1 - \lambda_L) = p_L(1 - \lambda_L)^2$ ($p_L > 0$).

Net income per representative higher-ranked worker:

$$y_H = m_H + \eta\lambda_H m_H - C_H(\lambda_H) - (1 - \lambda_H)m_H \\ + \kappa f_H(1 - \lambda_H)m_H + \rho_H(1 - \lambda_H),$$

with $C_H(\lambda_H) \approx b\eta\lambda_H m_H$ and $\rho_H(1 - \lambda_H) = p_H(1 - \lambda_H)^2$ ($p_H < 0$).

With linear utility $u(y) = y$, each class maximizes own net income (Nash, treating other's λ fixed).

2.1 Low-class first-order condition

$$\frac{\partial y_L}{\partial \lambda_L} \approx (\eta - b\eta - \kappa f_L + 1)m_L - 2p_L(1 - \lambda_L) = 0$$

Solving gives:

$$\lambda_L^* = 1 - \frac{m_L(\kappa f_L - \eta + b\eta - 1)}{2p_L},$$

bounded as $\lambda_L^* = \max(0, \min(1, \text{above}))$.

2.2 High-class first-order condition

$$\frac{\partial y_H}{\partial \lambda_H} \approx (\eta - b\eta - \kappa f_H + 1)m_H - 2p_H(1 - \lambda_H) = 0$$

Solving gives:

$$\lambda_H^* = 1 - \frac{m_H(\kappa f_H - \eta + b\eta - 1)}{2p_H},$$

bounded as $\lambda_H^* = \max(0, \min(1, \text{above}))$.

2.3 Interpretation with higher mass at bottom

When $f_L \gg f_H$, the aggregate lottery pool is dominated by lower-ranked contributions. The lottery becomes a “mass institution” primarily funded and used by the poor, while the small elite can safely allocate almost all to upskilling ($\lambda_H^* \approx 1$) without collapsing the pool. The model preserves churning at the bottom and stickiness at the top, now with realistic population asymmetry.